

Write your name here

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Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced

Paper 3: Statistics and Mechanics

Friday 15 June 2018 – Afternoon

Time: 2 hours

Paper Reference

9MA0/03

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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(1)

(1)

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2. Tessa owns a small clothes shop in a seaside town. She records the weekly sales figures, £ w , and the average weekly temperature, $t^{\circ}\text{C}$, for 8 weeks during the summer. The product moment correlation coefficient for these data is -0.915

- (a) Stating your hypotheses clearly and using a 5% level of significance, test whether or not the correlation between sales figures and average weekly temperature is negative. (3)
- (b) Suggest a possible reason for this correlation.

Tessa suggests that a linear regression model could be used to model these data.

- (c) State, giving a reason, whether or not the correlation coefficient is consistent with Tessa's suggestion. (1)
- (d) State, giving a reason, which variable would be the explanatory variable. (1)

Tessa calculated the linear regression equation as $w = 10\,755 - 171t$

- (e) Give an interpretation of the gradient of this regression equation. (1)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3. In an experiment a group of children each repeatedly throw a dart at a target. For each child, the random variable H represents the number of times the dart hits the target in the first 10 throws.

Peta models H as $B(10, 0.1)$

- (a) State two assumptions Peta needs to make to use her model.

(2)

- (b) Using Peta's model, find $P(H \geq 4)$

(1)

For each child the random variable F represents the number of the throw on which the dart first hits the target.

Using Peta's assumptions about this experiment,

- (c) find $P(F = 5)$

(2)

Thomas assumes that in this experiment no child will need more than 10 throws for the dart to hit the target for the first time. He models $P(F = n)$ as

$$P(F = n) = 0.01 + (n - 1) \times \alpha$$

where α is a constant.

- (d) Find the value of α

(4)

- (e) Using Thomas' model, find $P(F = 5)$

(1)

- (f) Explain how Peta's and Thomas' models differ in describing the probability that a dart hits the target in this experiment.

(1)



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 11 marks)



4. Charlie is studying the time it takes members of his company to travel to the office. He stands by the door to the office from 08 40 to 08 50 one morning and asks workers, as they arrive, how long their journey was.

(a) State the sampling method Charlie used.

(1)

(b) State and briefly describe an alternative method of non-random sampling Charlie could have used to obtain a sample of 40 workers.

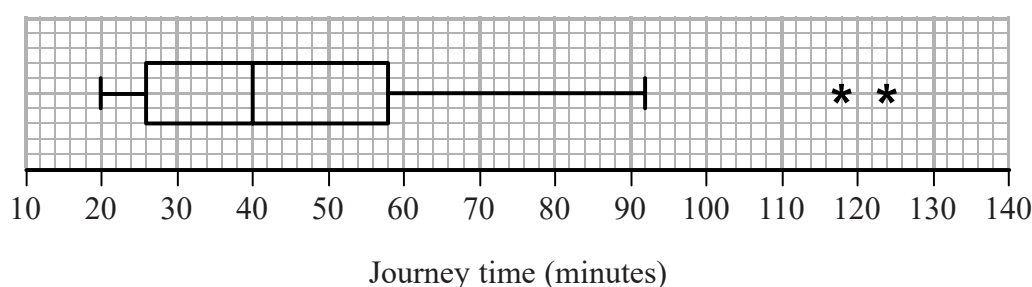
(2)

Taruni decided to ask every member of the company the time, x minutes, it takes them to travel to the office.

(c) State the data selection process Taruni used.

(1)

Taruni's results are summarised by the box plot and summary statistics below.



$$n = 95 \quad \sum x = 4133 \quad \sum x^2 = 202\,294$$

(d) Write down the interquartile range for these data.

(1)

(e) Calculate the mean and the standard deviation for these data.

(3)

(f) State, giving a reason, whether you would recommend using the mean and standard deviation or the median and interquartile range to describe these data.

(2)

Rana and David both work for the company and have both moved house since Taruni collected her data.

Rana's journey to work has changed from 75 minutes to 35 minutes and David's journey to work has changed from 60 minutes to 33 minutes.

Taruni drew her box plot again and only had to change two values.

(g) Explain which two values Taruni must have changed and whether each of these values has increased or decreased.

(3)



Question 4 continued

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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 13 marks)



5. The lifetime, L hours, of a battery has a normal distribution with mean 18 hours and standard deviation 4 hours.

Alice's calculator requires 4 batteries and will stop working when any one battery reaches the end of its lifetime.

- (a) Find the probability that a randomly selected battery will last for longer than 16 hours. (1)

At the start of her exams Alice put 4 new batteries in her calculator.
She has used her calculator for 16 hours, but has another 4 hours of exams to sit.

- (b) Find the probability that her calculator will not stop working for Alice's remaining exams. (5)

Alice only has 2 new batteries so, after the first 16 hours of her exams, although her calculator is still working, she randomly selects 2 of the batteries from her calculator and replaces these with the 2 new batteries.

- (c) Show that the probability that her calculator will not stop working for the remainder of her exams is 0.199 to 3 significant figures. (3)

After her exams, Alice believed that the lifetime of the batteries was more than 18 hours. She took a random sample of 20 of these batteries and found that their mean lifetime was 19.2 hours.

- (d) Stating your hypotheses clearly and using a 5% level of significance, test Alice's belief. (5)



Question 5 continued

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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 14 marks)

TOTAL FOR SECTION A IS 50 MARKS



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Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Friday 14 June 2019			
Afternoon		Paper Reference 9MA0-31	
Mathematics			
Advanced			
Paper 31: Statistics			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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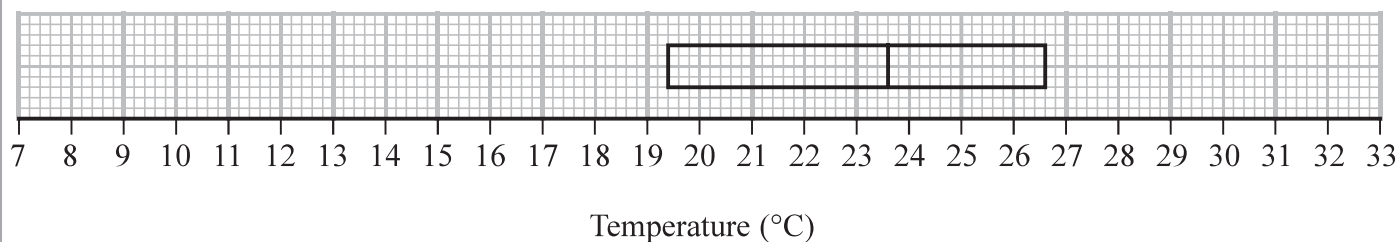
Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)



2.

**Figure 1**

The partially completed box plot in Figure 1 shows the distribution of daily mean air temperatures using the data from the large data set for Beijing in 2015

An outlier is defined as a value
 more than $1.5 \times \text{IQR}$ below Q_1 or
 more than $1.5 \times \text{IQR}$ above Q_3

The three lowest air temperatures in the data set are 7.6°C , 8.1°C and 9.1°C

The highest air temperature in the data set is 32.5°C

(a) Complete the box plot in Figure 1 showing clearly any outliers. (4)

(b) Using your knowledge of the large data set, suggest from which month the two outliers are likely to have come. (1)

Using the data from the large data set, Simon produced the following summary statistics for the daily mean air temperature, $x^\circ\text{C}$, for Beijing in 2015

$$n = 184 \quad \sum x = 4153.6 \quad S_{xx} = 4952.906$$

(c) Show that, to 3 significant figures, the standard deviation is 5.19°C (1)

Simon decides to model the air temperatures with the random variable

$$T \sim N(22.6, 5.19^2)$$

(d) Using Simon's model, calculate the 10th to 90th interpercentile range. (3)

Simon wants to model another variable from the large data set for Beijing using a normal distribution.

(e) State two variables from the large data set for Beijing that are **not** suitable to be modelled by a normal distribution. Give a reason for each answer. (2)



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Question 2 continued

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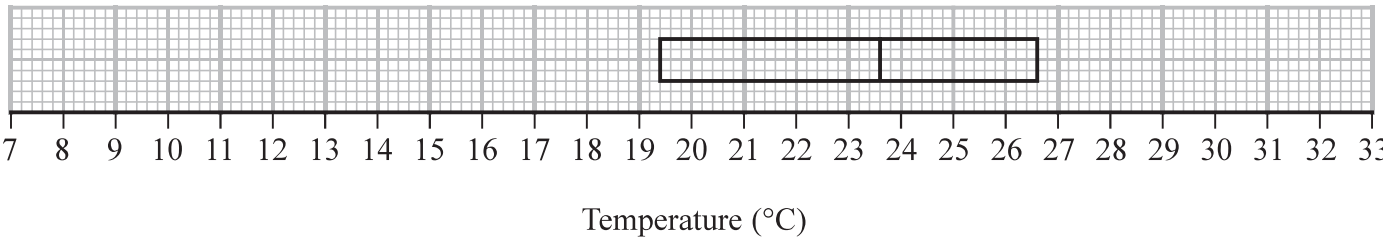


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Question 2 continued

Handwriting practice area with horizontal lines.

Only use this grid if you need to redraw your box plot.



(Total for Question 2 is 11 marks)



- Barbara is investigating the relationship between average income (GDP per capita), x US dollars, and average annual carbon dioxide (CO_2) emissions, y tonnes, for different countries.

She takes a random sample of 24 countries and finds the product moment correlation coefficient between average annual CO₂ emissions and average income to be 0.446

- (a) Stating your hypotheses clearly, test, at the 5% level of significance, whether or not the product moment correlation coefficient for all countries is greater than zero.

(3)

Barbara believes that a non-linear model would be a better fit to the data.

She codes the data using the coding $m = \log_{10}x$ and $c = \log_{10}y$ and obtains the model $c = -1.82 + 0.89m$

The product moment correlation coefficient between c and m is found to be 0.882

- (b) Explain how this value supports Barbara's belief.

(1)

- (c) Show that the relationship between y and x can be written in the form $y = ax^n$ where a and n are constants to be found.

(5)

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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4. Magali is studying the mean total cloud cover, in oktas, for Leuchars in 1987 using data from the large data set. The daily mean total cloud cover for all 184 days from the large data set is summarised in the table below.

Daily mean total cloud cover (oktas)	0	1	2	3	4	5	6	7	8
Frequency (number of days)	0	1	4	7	10	30	52	52	28

One of the 184 days is selected at random.

- (a) Find the probability that it has a daily mean total cloud cover of 6 or greater.

(1)

Magali is investigating whether the daily mean total cloud cover can be modelled using a binomial distribution.

She uses the random variable X to denote the daily mean total cloud cover and believes that $X \sim B(8, 0.76)$

Using Magali's model,

- (b) (i) find $P(X \geq 6)$

(2)

- (ii) find, to 1 decimal place, the expected number of days in a sample of 184 days with a daily mean total cloud cover of 7

(2)

- (c) Explain whether or not your answers to part (b) support the use of Magali's model.

(1)

There were 28 days that had a daily mean total cloud cover of 8

For these 28 days the daily mean total cloud cover for the **following** day is shown in the table below.

Daily mean total cloud cover (oktas)	0	1	2	3	4	5	6	7	8
Frequency (number of days)	0	0	1	1	2	1	5	9	9

- (d) Find the proportion of these days when the daily mean total cloud cover was 6 or greater.

(1)

- (e) Comment on Magali's model in light of your answer to part (d).

(2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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5. A machine puts liquid into bottles of perfume. The amount of liquid put into each bottle, D ml, follows a normal distribution with mean 25 ml

Given that 15% of bottles contain less than 24.63 ml

- (a) find, to 2 decimal places, the value of k such that $P(24.63 < D < k) = 0.45$ (5)

A random sample of 200 bottles is taken.

- (b) Using a normal approximation, find the probability that fewer than half of these bottles contain between 24.63 ml and k ml (3)

The machine is adjusted so that the standard deviation of the liquid put in the bottles is now 0.16 ml

Following the adjustments, Hannah believes that the mean amount of liquid put in each bottle is less than 25 ml

She takes a random sample of 20 bottles and finds the mean amount of liquid to be 24.94 ml

- (c) Test Hannah's belief at the 5% level of significance.
You should state your hypotheses clearly. (5)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 13 marks)

TOTAL FOR STATISTICS IS 50 MARKS



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Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Monday 19 October 2020			
Afternoon		Paper Reference 9MA0/31	
Mathematics Advanced Paper 31: Statistics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Information

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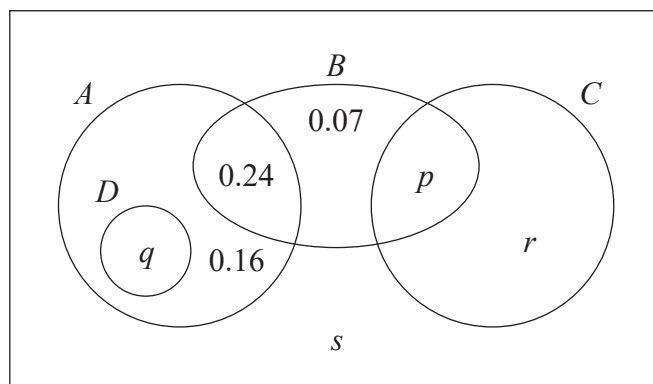
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1. The Venn diagram shows the probabilities associated with four events, A , B , C and D



- (a) Write down any pair of mutually exclusive events from A , B , C and D

(1)

Given that $P(B) = 0.4$

- (b) find the value of p

(1)

Given also that A and B are independent

- (c) find the value of q

(2)

Given further that $P(B'|C) = 0.64$

- (d) find

(i) the value of r

(ii) the value of s

(4)



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Question 1 continued

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(Total for Question 1 is 8 marks)



2. A random sample of 15 days is taken from the large data set for Perth in June and July 1987. The scatter diagram in Figure 1 displays the values of two of the variables for these 15 days.

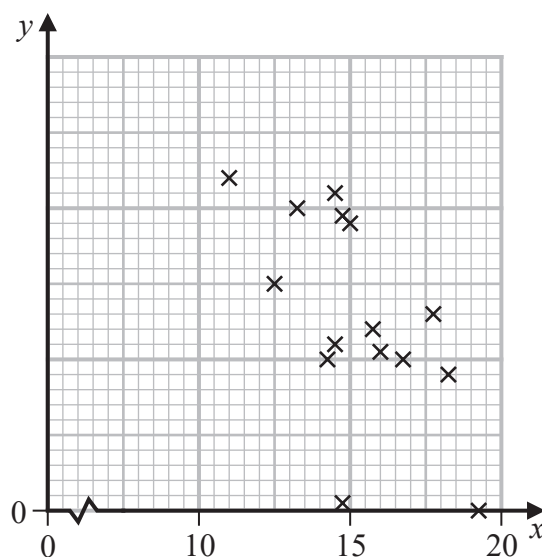


Figure 1

- (a) Describe the correlation.

(1)

The variable on the x -axis is Daily Mean Temperature measured in $^{\circ}\text{C}$.

- (b) Using your knowledge of the large data set,

- suggest which variable is on the y -axis,
- state the units that are used in the large data set for this variable.

(2)

Stav believes that there is a correlation between Daily Total Sunshine and Daily Maximum Relative Humidity at Heathrow.

He calculates the product moment correlation coefficient between these two variables for a random sample of 30 days and obtains $r = -0.377$

- (c) Carry out a suitable test to investigate Stav's belief at a 5% level of significance. State clearly

- your hypotheses
- your critical value

(3)

On a random day at Heathrow the Daily Maximum Relative Humidity was 97%

- (d) Comment on the number of hours of sunshine you would expect on that day, giving a reason for your answer.

(1)



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Question 2 continued

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Question 2 continued

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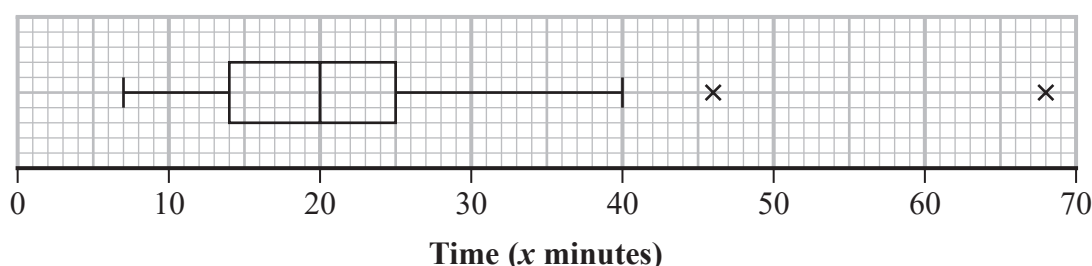
(Total for Question 2 is 7 marks)



3. Each member of a group of 27 people was timed when completing a puzzle.

The time taken, x minutes, for each member of the group was recorded.

These times are summarised in the following box and whisker plot.



- (a) Find the range of the times. (1)

- (b) Find the interquartile range of the times. (1)

For these 27 people $\sum x = 607.5$ and $\sum x^2 = 17\,623.25$

- (c) calculate the mean time taken to complete the puzzle, (1)

- (d) calculate the standard deviation of the times taken to complete the puzzle. (2)

Taruni defines an outlier as a value more than 3 standard deviations above the mean.

- (e) State how many outliers Taruni would say there are in these data, giving a reason for your answer. (1)

Adam and Beth also completed the puzzle in a minutes and b minutes respectively, where $a > b$.

When their times are included with the data of the other 27 people

- the median time increases
 - the mean time does not change
- (f) Suggest a possible value for a and a possible value for b , explaining how your values satisfy the above conditions. (3)
- (g) Without carrying out any further calculations, explain why the standard deviation of all 29 times will be lower than your answer to part (d). (1)



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Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 10 marks)



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4. The discrete random variable D has the following probability distribution

d	10	20	30	40	50
$P(D = d)$	$\frac{k}{10}$	$\frac{k}{20}$	$\frac{k}{30}$	$\frac{k}{40}$	$\frac{k}{50}$

where k is a constant.

- (a) Show that the value of k is $\frac{600}{137}$ (2)

The random variables D_1 and D_2 are independent and each have the same distribution as D .

- (b) Find $P(D_1 + D_2 = 80)$
Give your answer to 3 significant figures.
- (3)**

A single observation of D is made.

The value obtained, d , is the common difference of an arithmetic sequence.

The first 4 terms of this arithmetic sequence are the angles, measured in degrees, of quadrilateral Q

- (c) Find the exact probability that the smallest angle of Q is more than 50° (5)



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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 10 marks)



5. A health centre claims that the time a doctor spends with a patient can be modelled by a normal distribution with a mean of 10 minutes and a standard deviation of 4 minutes.
- (a) Using this model, find the probability that the time spent with a randomly selected patient is more than 15 minutes. (1)

Some patients complain that the mean time the doctor spends with a patient is more than 10 minutes.

The receptionist takes a random sample of 20 patients and finds that the mean time the doctor spends with a patient is 11.5 minutes.

- (b) Stating your hypotheses clearly and using a 5% significance level, test whether or not there is evidence to support the patients' complaint. (4)

The health centre also claims that the time a dentist spends with a patient during a routine appointment, T minutes, can be modelled by the normal distribution where $T \sim N(5, 3.5^2)$

- (c) Using this model,
- (i) find the probability that a routine appointment with the dentist takes less than 2 minutes (1)
- (ii) find $P(T < 2 \mid T > 0)$ (3)
- (iii) hence explain why this normal distribution may not be a good model for T . (1)

The dentist believes that she cannot complete a routine appointment in less than 2 minutes.

She suggests that the health centre should use a refined model only including values of $T > 2$

- (d) Find the median time for a routine appointment using this new model, giving your answer correct to one decimal place. (5)



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Question 5 continued

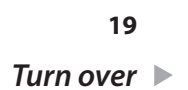
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Question 5 continued

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(Total for Question 5 is 15 marks)

TOTAL FOR STATISTICS IS 50 MARKS



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Pearson Edexcel Level 3 GCE

Monday 18 October 2021 – Afternoon

Paper reference	9MA0/31
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Mathematics

Advanced

PAPER 31: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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- The total mark for this part of the examination is 50. There are 6 questions.
- The marks for **each** question are shown in brackets
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Advice

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Turn over ►

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1. (a) State one disadvantage of using quota sampling compared with simple random sampling.

(1)

In a university 8% of students are members of the university dance club.

A random sample of 36 students is taken from the university.

The random variable X represents the number of these students who are members of the dance club.

- (b) Using a suitable model for X , find

(i) $P(X = 4)$

(ii) $P(X \geq 7)$

(3)

Only 40% of the university dance club members can dance the tango.

- (c) Find the probability that a student is a member of the university dance club and can dance the tango.

(1)

A random sample of 50 students is taken from the university.

- (d) Find the probability that fewer than 3 of these students are members of the university dance club and can dance the tango.

(2)



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Question 1 continued.

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



2. Marc took a random sample of 16 students from a school and for each student recorded

- the number of letters, x , in their last name
- the number of letters, y , in their first name

His results are shown in the scatter diagram on the next page.

(a) Describe the correlation between x and y .

(1)

Marc suggests that parents with long last names tend to give their children shorter first names.

(b) Using the scatter diagram comment on Marc's suggestion, giving a reason for your answer.

(1)

The results from Marc's random sample of 16 observations are given in the table below.

x	3	6	8	7	5	3	11	3	4	5	4	9	7	10	6	6
y	7	7	4	4	6	8	5	5	8	4	7	4	5	5	6	3

(c) Use your calculator to find the product moment correlation coefficient between x and y for these data.

(1)

(d) Test whether or not there is evidence of a negative correlation between the number of letters in the last name and the number of letters in the first name.

You should

- state your hypotheses clearly
- use a 5% level of significance

(3)

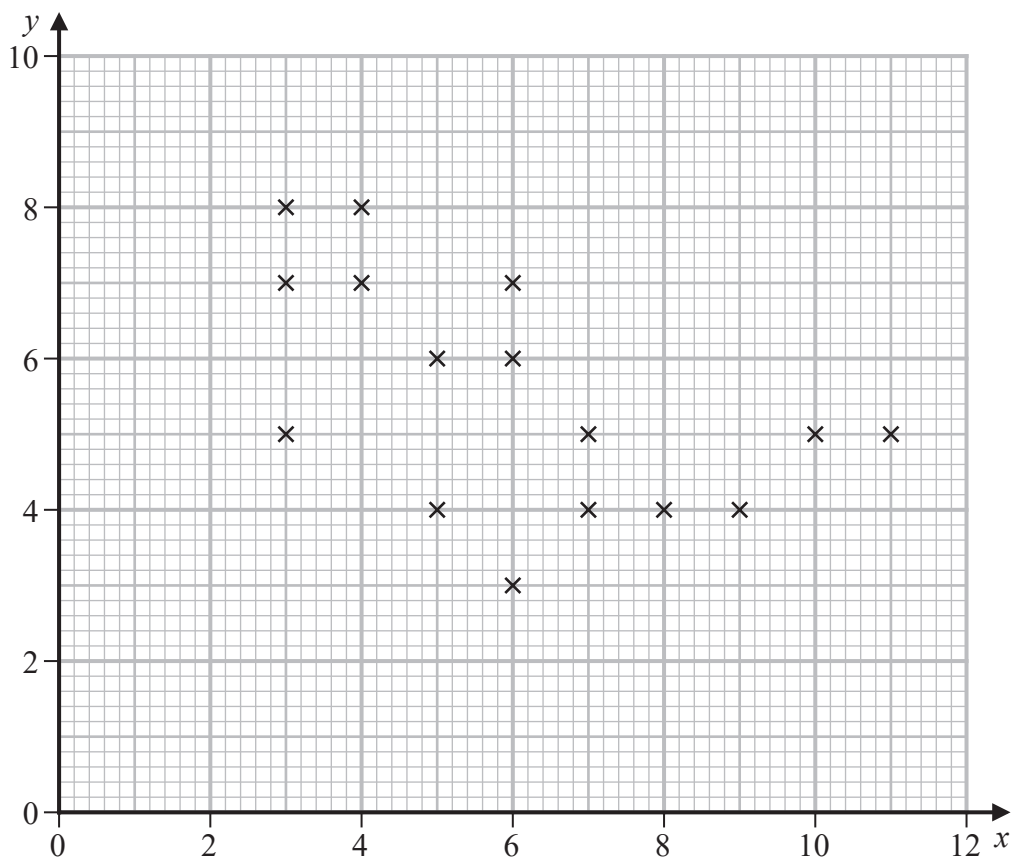


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Question 2 continued.



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Question 2 continued.

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3. Stav is studying the large data set for September 2015

He codes the variable Daily Mean Pressure, x , using the formula $y = x - 1010$

The data for all 30 days from Hurn are summarised by

$$\sum y = 214 \quad \sum y^2 = 5912$$

- (a) State the units of the variable x (1)
- (b) Find the mean Daily Mean Pressure for these 30 days. (2)
- (c) Find the standard deviation of Daily Mean Pressure for these 30 days. (3)

Stav knows that, in the UK, winds circulate

- in a **clockwise** direction around a region of **high** pressure
- in an **anticlockwise** direction around a region of **low** pressure

The table gives the Daily Mean Pressure for 3 locations from the large data set on 26/09/2015

Location	Heathrow	Hurn	Leuchars
Daily Mean Pressure	1029	1028	1028
Cardinal Wind Direction			

The Cardinal Wind Directions for these 3 locations on 26/09/2015 were, in random order,

W NE E

You may assume that these 3 locations were under a single region of pressure.

- (d) Using your knowledge of the large data set, place each of these Cardinal Wind Directions in the correct location in the table. (2)
- Give a reason for your answer.



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Question 3 continued.

Lined area for writing the answer to Question 3.

(Total for Question 3 is 8 marks)



P 6 8 8 2 8 A 0 9 2 0

4. A large college produces three magazines.

One magazine is about green issues, one is about equality and one is about sports.

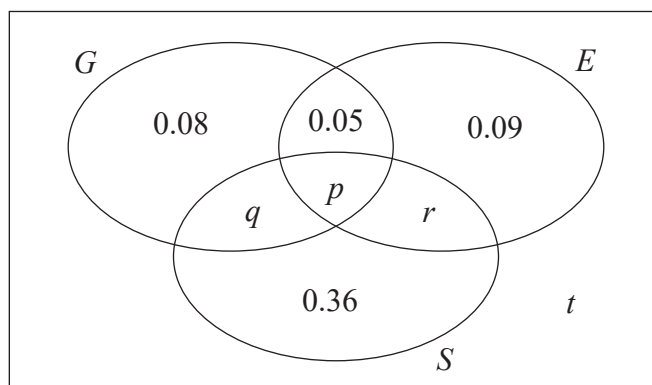
A student at the college is selected at random and the events G , E and S are defined as follows

G is the event that the student reads the magazine about green issues

E is the event that the student reads the magazine about equality

S is the event that the student reads the magazine about sports

The Venn diagram, where p , q , r and t are probabilities, gives the probability for each subset.



- (a) Find the proportion of students in the college who read exactly one of these magazines.

(1)

No students read all three magazines and $P(G) = 0.25$

- (b) Find

(i) the value of p

(ii) the value of q

(3)

Given that $P(S | E) = \frac{5}{12}$

- (c) find

(i) the value of r

(ii) the value of t

(4)

- (d) Determine whether or not the events $(S \cap E')$ and G are independent. Show your working clearly.

(3)



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Question 4 continued.

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Question 4 continued.

(Total for Question 4 is 11 marks)



5. The heights of females from a country are normally distributed with
- a mean of 166.5 cm
 - a standard deviation of 6.1 cm

Given that 1% of females from this country are shorter than k cm,

- (a) find the value of k (2)
- (b) Find the proportion of females from this country with heights between 150 cm and 175 cm (1)

A female, from this country, is chosen at random from those with heights between 150 cm and 175 cm

- (c) Find the probability that her height is more than 160 cm (4)

The heights of females from a different country are normally distributed with a standard deviation of 7.4 cm

Mia believes that the mean height of females from this country is less than 166.5 cm

Mia takes a random sample of 50 females from this country and finds the mean of her sample is 164.6 cm

- (d) Carry out a suitable test to assess Mia's belief.
You should
- state your hypotheses clearly
 - use a 5% level of significance
- (4)**

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Question 5 continued.

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Question 5 continued.

(Total for Question 5 is 11 marks)



6. The discrete random variable X has the following probability distribution

x	a	b	c
$P(X = x)$	$\log_{36} a$	$\log_{36} b$	$\log_{36} c$

where

- a, b and c are distinct integers ($a < b < c$)
- all the probabilities are greater than zero

(a) Find

- the value of a
- the value of b
- the value of c

Show your working clearly.

(5)

The independent random variables X_1 and X_2 each have the same distribution as X

(b) Find $P(X_1 = X_2)$

(2)



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Question 6 continued.

Lined area for writing the answer to Question 6.



Question 6 continued.

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(Total for Question 6 is 7 marks)

TOTAL FOR STATISTICS IS 50 MARKS



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Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Paper reference	9MA0/31
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Mathematics

Advanced

PAPER 31: Statistics

You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks
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Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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Turn over ►

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- (3)

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Question 1 continued

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(Total for Question 1 is 6 marks)



2. A manufacturer uses a machine to make metal rods.

The length of a metal rod, L cm, is normally distributed with

- a mean of 8 cm
- a standard deviation of x cm

Given that the proportion of metal rods less than 7.902 cm in length is 2.5%

(a) show that $x = 0.05$ to 2 decimal places.

(2)

(b) Calculate the proportion of metal rods that are between 7.94 cm and 8.09 cm in length.

(1)

The **cost** of producing a single metal rod is 20p

A metal rod

- where $L < 7.94$ is **sold** for scrap for 5p
- where $7.94 \leq L \leq 8.09$ is **sold** for 50p
- where $L > 8.09$ is shortened for an extra **cost** of 10p and then **sold** for 50p

(c) Calculate the expected profit per 500 of the metal rods.
Give your answer to the nearest pound.

(5)

The same manufacturer makes metal hinges in large batches.

The hinges each have a probability of 0.015 of having a fault.

A random sample of 200 hinges is taken from each batch and the batch is accepted if fewer than 6 hinges are faulty.

The manufacturer's aim is for 95% of batches to be accepted.

(d) Explain whether the manufacturer is likely to achieve its aim.

(4)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 12 marks)



3. Dian uses the large data set to investigate the Daily Total Rainfall, r mm, for Camborne.

(a) Write down how a value of $0 < r \leq 0.05$ is recorded in the large data set.

(1)

Dian uses the data for the 31 days of August 2015 for Camborne and calculates the following statistics

$$n = 31 \quad \sum r = 174.9 \quad \sum r^2 = 3523.283$$

(b) Use these statistics to calculate

(i) the mean of the Daily Total Rainfall in Camborne for August 2015,

(ii) the standard deviation of the Daily Total Rainfall in Camborne for August 2015.

(3)

Dian believes that the mean Daily Total Rainfall in August is less in the South of the UK than in the North of the UK.

The mean Daily Total Rainfall in Leuchars for August 2015 is 1.72 mm to 2 decimal places.

(c) State, giving a reason, whether this provides evidence to support Dian's belief.

(2)

Dian uses the large data set to estimate the proportion of days with no rain in Camborne for 1987 to be 0.27 to 2 decimal places.

(d) Explain why the distribution $B(14, 0.27)$ might **not** be a reasonable model for the number of days without rain for a 14-day summer event.

(1)



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Question 3 continued

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(Total for Question 3 is 7 marks)



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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 6 marks)



5. A company has 1825 employees.
The employees are classified as professional, skilled or elementary.

The following table shows

- the number of employees in each classification
- the two areas, A or B , where the employees live

	A	B
Professional	740	380
Skilled	275	90
Elementary	260	80

An employee is chosen at random.

Find the probability that this employee

(a) is skilled, (1)

(b) lives in area B and is not a professional. (1)

Some classifications of employees are more likely to work from home.

- 65% of professional employees in both area A and area B work from home
- 40% of skilled employees in both area A and area B work from home
- 5% of elementary employees in both area A and area B work from home
- Event F is that the employee is a professional
- Event H is that the employee works from home
- Event R is that the employee is from area A

(c) Using this information, complete the Venn diagram on the opposite page. (4)

(d) Find $P(R' \cap F)$ (1)

(e) Find $P([H \cup R]')$ (1)

(f) Find $P(F | H)$ (2)



Question 5 continued

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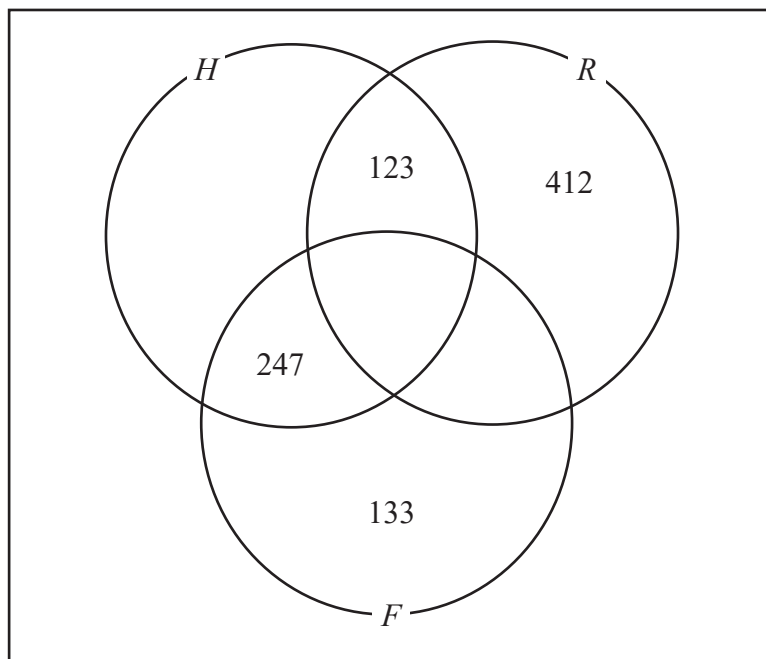
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Question 5 continued

Only use this diagram if you need to redraw your Venn diagram.

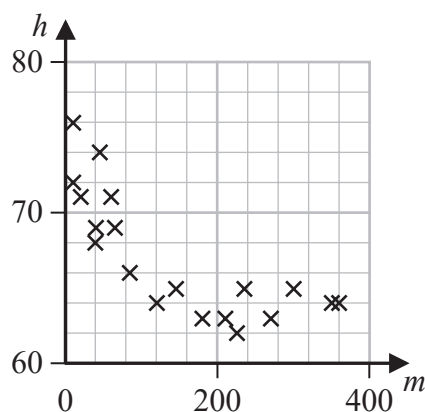


(Total for Question 5 is 10 marks)

6. Anna is investigating the relationship between exercise and resting heart rate. She takes a random sample of 19 people in her year at school and records for each person

- their resting heart rate, h beats per minute
- the number of minutes, m , spent exercising each week

Her results are shown on the scatter diagram.



- (a) Interpret the nature of the relationship between h and m

(1)

Anna codes the data using the formulae

$$x = \log_{10} m$$

$$y = \log_{10} h$$

The product moment correlation coefficient between x and y is -0.897

- (b) Test whether or not there is significant evidence of a negative correlation between x and y

You should

- state your hypotheses clearly
- use a 5% level of significance
- state the critical value used

(3)

The equation of the line of best fit of y on x is

$$y = -0.05x + 1.92$$

- (c) Use the equation of the line of best fit of y on x to find a model for h on m in the form

$$h = am^k$$

where a and k are constants to be found.

(5)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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(Total for Question 6 is 9 marks)

TOTAL FOR STATISTICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Tuesday 20 June 2023

Afternoon

Paper reference **9MA0/31**

Mathematics

Advanced

PAPER 31: Statistics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Turn over ►

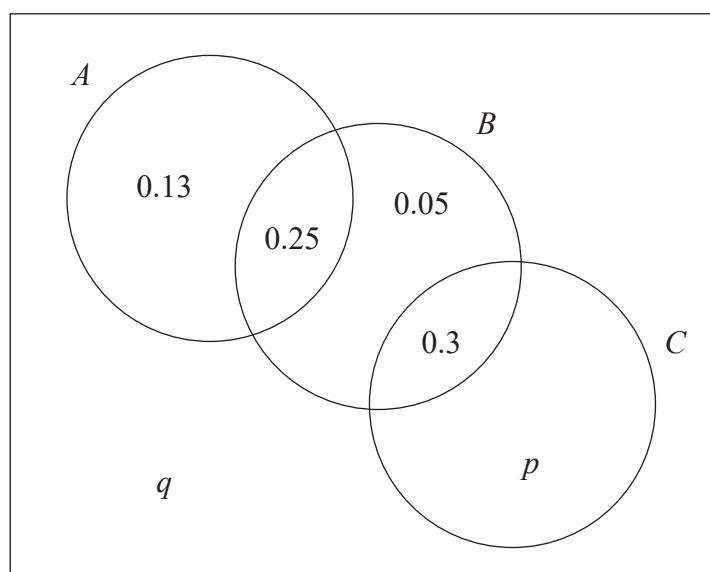
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1. The Venn diagram, where p and q are probabilities, shows the three events A , B and C and their associated probabilities.



- (a) Find $P(A)$ (1)

The events B and C are independent.

- (b) Find the value of p and the value of q (3)

- (c) Find $P(A|B')$ (2)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2. A machine fills packets with sweets and $\frac{1}{7}$ of the packets also contain a prize.

The packets of sweets are placed in boxes before being delivered to shops.
There are 40 packets of sweets in each box.

The random variable T represents the number of packets of sweets that contain a prize in each box.

- (a) State a condition needed for T to be modelled by $B(40, \frac{1}{7})$ (1)

A box is selected at random.

- (b) Using $T \sim B(40, \frac{1}{7})$ find (2)
- (i) the probability that the box has exactly 6 packets containing a prize,
 - (ii) the probability that the box has fewer than 3 packets containing a prize.

Kamil's sweet shop buys 5 boxes of these sweets.

- (c) Find the probability that exactly 2 of these 5 boxes have fewer than 3 packets containing a prize. (2)

Kamil claims that the proportion of packets containing a prize is less than $\frac{1}{7}$

A random sample of 110 packets is taken and 9 packets contain a prize.

- (d) Use a suitable test to assess Kamil's claim. (4)
- You should
- state your hypotheses clearly
 - use a 5% level of significance



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Question 2 continued

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Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 9 marks)



3. Ben is studying the Daily Total Rainfall, x mm, in Leeming for 1987

He used all the data from the large data set and summarised the information in the following table.

x	0	0.1–0.5	0.6–1.0	1.1–1.9	2.0–4.0	4.1–6.9	7.0–12.0	12.1–20.9	21.0–32.0	tr
Frequency	55	18	18	21	17	9	9	6	2	29

- (a) Explain how the data will need to be cleaned before Ben can start to calculate statistics such as the mean and standard deviation.

(2)

Using all 184 of these values, Ben estimates $\sum x = 390$ and $\sum x^2 = 4336$

- (b) Calculate estimates for

- (i) the mean Daily Total Rainfall,
(ii) the standard deviation of the Daily Total Rainfall.

(3)

Ben suggests using the statistic calculated in part (b)(i) to estimate the annual mean Daily Total Rainfall in Leeming for 1987

- (c) Using your knowledge of the large data set,

- (i) give a reason why these data would not be suitable,
(ii) state, giving a reason, how you would expect the estimate in part (b)(i) to differ from the actual annual mean Daily Total Rainfall in Leeming for 1987

(2)



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Question 3 continued

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(Total for Question 3 is 7 marks)



4. A study was made of adult men from region A of a country. It was found that their heights were normally distributed with a mean of 175.4 cm and standard deviation 6.8 cm.

(a) Find the proportion of these men that are taller than 180 cm.

(1)

A student claimed that the mean height of adult men from region B of this country was different from the mean height of adult men from region A .

A random sample of 52 adult men from region B had a mean height of 177.2 cm.

The student assumed that the standard deviation of heights of adult men was 6.8 cm both for region A and region B .

(b) Use a suitable test to assess the student's claim.

You should

- state your hypotheses clearly
- use a 5% level of significance

(4)

(c) Find the p -value for the test in part (b)

(1)



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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 6 marks)



5. Tisam is playing a game.
She uses a ball, a cup and a spinner.

The random variable X represents the number the spinner lands on when it is spun.
The probability distribution of X is given in the following table

x	20	50	80	100
$P(X = x)$	a	b	c	d

where a , b , c and d are probabilities.

To play the game

- the spinner is spun to obtain a value of x
- Tisam then stands x cm from the cup and tries to throw the ball into the cup

The event S represents the event that Tisam successfully throws the ball into the cup.

To model this game Tisam assumes that

- $P(S | \{X = x\}) = \frac{k}{x}$ where k is a constant
- $P(S \cap \{X = x\})$ should be the same whatever value of x is obtained from the spinner

Using Tisam's model,

(a) show that $c = \frac{8}{5}b$ (2)

(b) find the probability distribution of X (5)

Nav tries, a large number of times, to throw the ball into the cup from a distance of 100 cm.

He successfully gets the ball in the cup 30% of the time.

- (c) State, giving a reason, why Tisam's model of this game is not suitable to describe Nav playing the game for all values of X (1)



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Question 5 continued

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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 8 marks)



6. A medical researcher is studying the number of hours, T , a patient stays in hospital following a particular operation.

The histogram on the page opposite summarises the results for a random sample of 90 patients.

- (a) Use the histogram to estimate $P(10 < T < 30)$ (2)

For these 90 patients the time spent in hospital following the operation had

- a mean of 14.9 hours
- a standard deviation of 9.3 hours

Tomas suggests that T can be modelled by $N(14.9, 9.3^2)$

- (b) With reference to the histogram, state, giving a reason, whether or not Tomas' model could be suitable. (1)

Xiang suggests that the frequency polygon based on this histogram could be modelled by a curve with equation

$$y = kxe^{-x} \quad 0 \leq x \leq 4$$

where

- x is measured in **tens of hours**
- k is a constant

- (c) Use algebraic integration to show that

$$\int_0^n xe^{-x} dx = 1 - (n+1)e^{-n} \quad (4)$$

- (d) Show that, for Xiang's model, $k = 99$ to the nearest integer. (3)

- (e) Estimate $P(10 < T < 30)$ using

- (i) Tomas' model of $T \sim N(14.9, 9.3^2)$ (1)

- (ii) Xiang's curve with equation $y = 99xe^{-x}$ and the answer to part (c) (2)

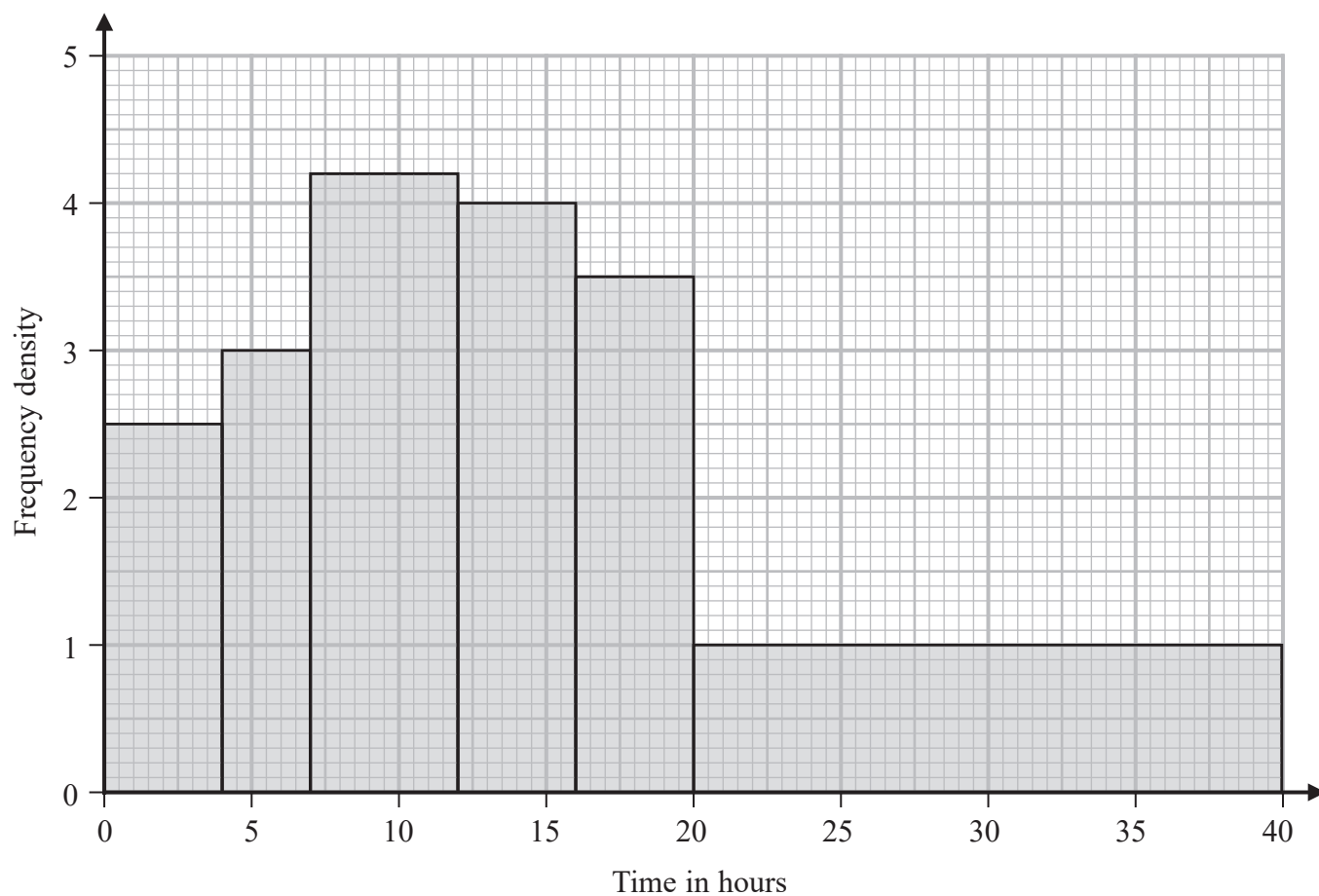
The researcher decides to use Xiang's curve to model $P(a < T < b)$

- (f) State one limitation of Xiang's model. (1)



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Question 6 continued



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Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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(Total for Question 6 is 14 marks)

TOTAL FOR STATISTICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	
Pearson Edexcel Level 3 GCE			
Thursday 20 June 2024			
Afternoon		Paper reference	9MA0/31
Mathematics Advanced PAPER 31: Statistics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

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Turn over ►

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1. Xian rolls a fair die 10 times.

The random variable X represents the number of times the die lands on a six.

- (a) Using a suitable distribution for X , find

(i) $P(X = 3)$

(ii) $P(X < 3)$

(3)

Xian repeats this experiment each day for 60 days and records the number of days when $X = 3$

- (b) Find the probability that there were at least 12 days when $X = 3$

(3)

- (c) Find an estimate for the total number of sixes that Xian will roll during these 60 days.

(1)

- (d) Use a normal approximation to estimate the probability that Xian rolls a total of more than 95 sixes during these 60 days.

(4)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 11 marks)



2. Amar is studying the flight of a bird from its nest.

He measures the bird's height above the ground, h metres, at time t seconds for 10 values of t

Amar finds the equation of the regression line for the data to be $h = 38.6 - 1.28t$

- (a) Interpret the gradient of this line.

(1)

The product moment correlation coefficient between h and t is -0.510

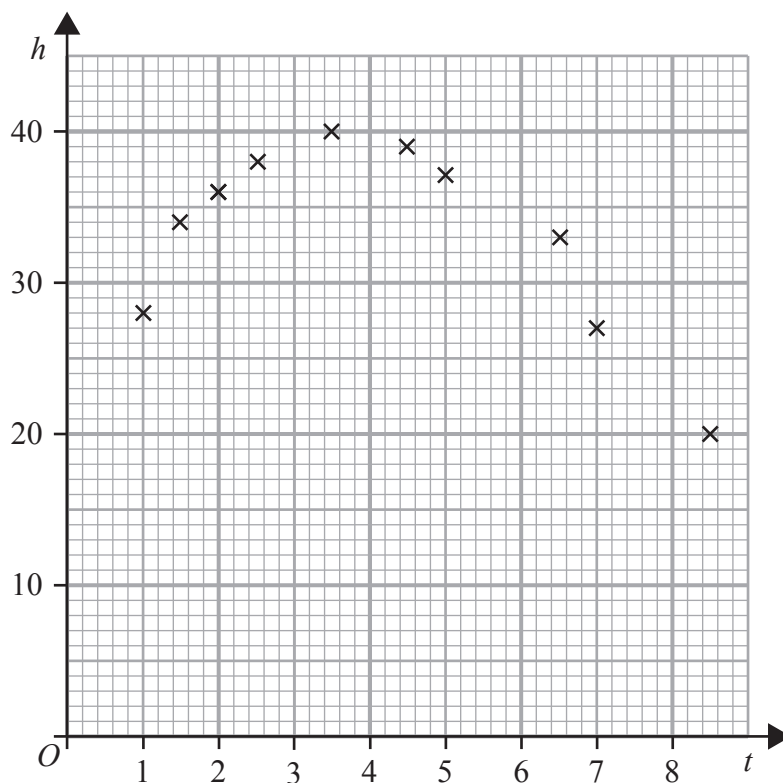
- (b) Test whether or not there is evidence of a negative correlation between the height above the ground and the time during the flight.

You should

- state your hypotheses clearly
- use a 5% level of significance
- state the critical value used

(3)

Jane draws the following scatter diagram for Amar's data.



- (c) With reference to the scatter diagram, state, giving a reason, whether or not the regression line $h = 38.6 - 1.28t$ is an appropriate model for these data.

(1)

Jane suggests an improved model using the variable $u = (t - k)^2$ where k is a constant.

She obtains the equation $h = 38.1 - 0.78u$

- (d) Choose a suitable value for k to write Jane's improved model for h in terms of t only.

(1)



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Question 2 continued

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Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



3. Ming is studying the large data set for Perth in 2015

He intended to use all the data available to find summary statistics for the Daily Mean Air Temperature, x °C.

Unfortunately, Ming selected an incorrect variable on the spreadsheet.

This incorrect variable gave a mean of 5.3 and a standard deviation of 12.4

- (a) Using your knowledge of the large data set, suggest which variable Ming selected. (1)

The correct values for the Daily Mean Air Temperature are summarised as

$$n = 184 \qquad \sum x = 2801.2 \qquad \sum x^2 = 44\,695.4$$

- (b) Calculate the mean and standard deviation for these data. (3)

One of the months from the large data set for Perth in 2015 has

- mean $\bar{x} = 19.4$
- standard deviation $\sigma_x = 2.83$

for Daily Mean Air Temperature.

- (c) Suggest, giving a reason, a month these data may have come from. (2)

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)



4. The proportion of left-handed adults in a country is 10%
 Freya believes that the proportion of left-handed adults under the age of 25 in this country is different from 10%
 She takes a random sample of 40 adults under the age of 25 from this country to investigate her belief.

- (a) Find the critical region for a suitable test to assess Freya's belief.

You should

- state your hypotheses clearly
- use a 5% level of significance
- state the probability of rejection in each tail

(4)

- (b) Write down the actual significance level of your test in part (a)

(1)

In Freya's sample 7 adults were left-handed.

- (c) With reference to your answer in part (a) comment on Freya's belief.

(1)

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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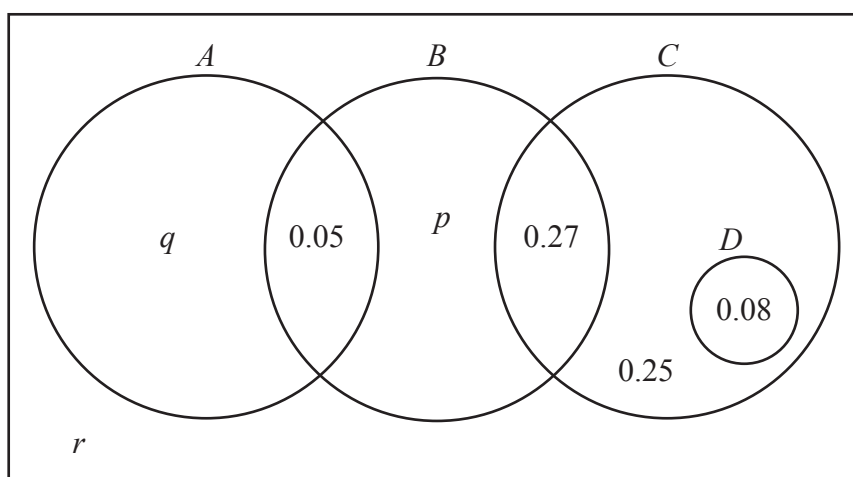
Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



6. The Venn diagram, where p , q and r are probabilities, shows the events A , B , C and D and associated probabilities.



- (a) State any pair of mutually exclusive events from A , B , C and D (1)

The events B and C are independent.

- (b) Find the value of p (2)

- (c) Find the greatest possible value of $P(A | B')$ (3)

Given that $P(B | A') = 0.5$

- (d) find the value of q and the value of r (3)

- (e) Find $P([A \cup B]' \cap C)$ (1)

- (f) Use set notation to write an expression for the event with probability p (1)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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(Total for Question 6 is 11 marks)

TOTAL FOR STATISTICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Thursday 19 June 2025

Afternoon	Paper reference	9MA0/31
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Mathematics

Advanced

PAPER 31: Statistics

You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks
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Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 6 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Bag A contains 5 red, 4 yellow and 3 green beads.

Bag B only contains red and yellow beads.

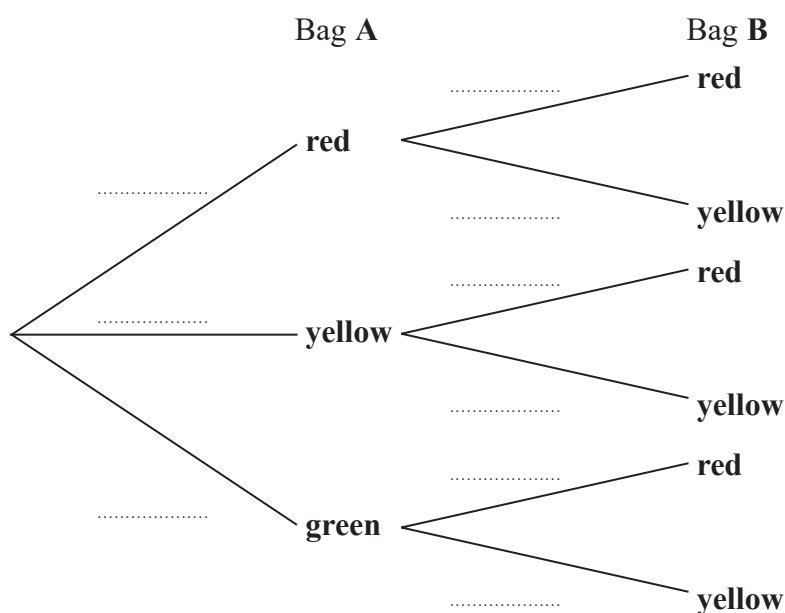
A bead is selected at random from bag A and a second bead is selected at random from bag B

Given that the probability that both beads selected are yellow is $\frac{3}{16}$

- (a) (i) find the probability of selecting a yellow bead from bag B

(2)

- (ii) Hence complete the tree diagram below by finding the probability on each branch.



(2)

- (b) Find the exact probability that at least one of the two beads selected is yellow.

(3)

The event X is that at least one of the beads selected is yellow.

The event W is that a green bead is selected.

- (c) Find the exact value of $P(W|X)$

(2)



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Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 9 marks)



- (c) State, giving a reason, which coach is more likely to have trained the winner. (2)

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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 5 marks)



3. A manufacturer makes components for the car industry.

On average 1.5% of the components made are defective.

(a) Using a suitable model, find the probability that in a random sample of 36 components there are

(i) exactly 2 defective components,

(ii) more than 3 defective components.

(3)

Once every six months, the manufacturer tests whether the proportion of defective components has changed using hypotheses $H_0: p = 0.015$ and $H_1: p \neq 0.015$

A random sample of 260 components is taken and 8 are found to be defective.

(b) Using a 5% level of significance, complete the test.

(3)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 6 marks)



4. Kay is studying the variables Daily Total Sunshine (x) and Daily Total Rainfall (y) from the large data set for Leeming in 2015

Kay starts with 5th May and then selects every 10th day thereafter.

- (a) State the name of the sampling technique Kay uses. (1)

Kay wants to find the regression line of y on x for these data.

- (b) Using your knowledge of the large data set, explain how Kay might need to clean these data before finding the equation of the regression line. (1)

The equation of the regression line Kay finds is $y = 0.741 + 0.199x$

- (c) Using your knowledge of the large data set,
(i) state the units of the gradient of the regression line,
(ii) give an interpretation of the y -intercept of the regression line. (2)

Kay's teacher claimed that the greater the amount of sunshine in a day the lower the amount of rain there should be.

- (d) State, giving a reason, whether the teacher's claim is true for Kay's data. (1)

The teacher used all the data for these variables from the large data set for Leeming in 2015, as a sample.

The teacher calculated the product moment correlation coefficient for x and y to be -0.160

In a suitable test to determine whether there is evidence to support the teacher's claim, the p -value was 0.015

- (e) Using a 5% level of significance, state the hypotheses and conclusion for this test. (2)

- (f) For the test in part (e) describe
(i) the sample,
(ii) a possible population. (2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 9 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

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(Total for Question 5 is 10 marks)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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(Total for Question 6 is 11 marks)

TOTAL FOR STATISTICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mock Paper Set 1

Paper Reference **9MA0-31**

Mathematics

Advanced

Paper 31: Statistics

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- Answer the questions in the spaces provided
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- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

Answer ALL questions. Write your answers in the spaces provided.

1. The daily mean air temperatures from the large data set, $x^{\circ}\text{C}$, for the month of June 2015 in Jacksonville are summarised in the table below.

Daily mean air temperature ($^{\circ}\text{C}$)	$22 \leq x < 24$	$24 \leq x < 25$	$25 \leq x < 26$	$26 \leq x < 27$	$27 \leq x < 28$	$28 \leq x < 32$
Frequency	2	5	7	4	6	6

- (a) Use your calculator to estimate the mean and the standard deviation of the daily mean air temperatures from the large data set, for the month of June 2015 in Jacksonville.

Give each of your answers to 3 significant figures.

(2)

The mean and standard deviation for the daily mean air temperatures from the large data set for Perth in June 2015 are 14.8°C and 2.37°C respectively.

The minimum daily mean air temperature in Perth in June 2015 was 8.8°C and the maximum daily mean air temperature was 18.5°C

- (b) Using limits for outliers of

$$\text{mean} - 3 \times \text{standard deviation}$$

$$\text{mean} + 3 \times \text{standard deviation}$$

show that there are no outliers in the data for Perth in June 2015.

(2)

- (c) (i) Assuming each location is typical of the hemisphere it is in, suggest what these means and standard deviations imply about the relative daily mean air temperature in June 2015 in each hemisphere.

Give reasons for your answers.

(2)

- (ii) Comment on the validity of the assumption in (i)

(1)

Amy models the daily mean air temperature in summer in Jacksonville by $N(27, 2.1^2)$

A survey found that the typical British person says that 29°C or above is 'too hot'.

A random sample of 30 summer days in Jacksonville is taken.

- (d) Use Amy's model to predict the number of these days when the mean air temperature would be considered 'too hot' for a typical British person visiting Jacksonville.

(2)



Question 1 continued

Lined area for writing the answer to Question 1.

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Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 9 marks)



2. An ornithologist believes that there is a relationship between the tail length, t mm, and the wing length, w mm, of female hook-billed kites. A random sample of size 10 is taken from a database of these kites and the relevant data is given in the table below.

$t(\text{mm})$	191	197	208	180	188	210	196	191	179	208
$w(\text{mm})$	284	285	288	273	280	283	288	271	257	289

The ornithologist plans to use a linear regression model based on these data and interpolate or extrapolate as necessary to estimate the wing length of other female hook-billed kites from their tail length.

- (a) (i) Explain what is meant by extrapolation. (1)
- (ii) Explain the dangers of extrapolation. (1)

The ornithologist attempts to calculate the product moment correlation coefficient, r , and obtains a value of 1.3

- (b) Explain how she should be able to identify that this is incorrect without carrying out any further calculations. (1)
- (c) Use your calculator to find the correct value of the product moment correlation coefficient, r . (1)
- (d) Stating your hypotheses clearly test, at the 1% significance level, whether or not there is evidence that the product moment correlation coefficient for the population is positive. (3)
- (e) Explain what your test in part (d) suggests about female hook-billed kites. (1)



Question 2 continued

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Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. A company maintains machines.

It has two types of contract, a service contract and a repair contract.

The company classes its customers as new customers or existing customers.

The table gives information about the company's customers.

	Service contract	Repair contract
New customer	65	82
Existing customer	231	262

The company is going to survey its customers. It plans to take a sample of 100 of its customers, stratified by customer type and contract type.

(a) Work out how many new customers with repair contracts should be sampled.

(2)

The company has developed a test for a certain fault in the machines it services. The test sometimes gives incorrect results.

The company collects information from a sample of randomly selected machines.

- 2% of the machines have the fault
- 70% of the machines with the fault test positive for the fault
- 10% of the machines without the fault test positive for the fault.

A machine is selected at random from the sample of the machines, and tests positive for the fault.

(b) (i) Calculate the probability that the machine has the fault.

(4)

- (ii) Comment on the usefulness of the company's test.
Give a reason for your answer.

(1)

When the company services the machines, it checks two components, α and β , for wear and tear and replaces these if needed.

Event A is that component α needs to be replaced.

Event B is that component β needs to be replaced.

The probability that component α needs to be replaced is 0.35

The probability that component β needs to be replaced is 0.55

The probability that neither component needs to be replaced is 0.28

(c) Show that events A and B are not independent.

(2)

(d) Find the probability that component α or component β needs to be replaced, but not both.

(2)



Question 3 continued

Lined area for writing the answer to Question 3 continued.

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Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 11 marks)



4. A company has a customer services call centre. The company believes that the time taken to complete a call to the call centre may be modelled by a normal distribution with mean 16 minutes and standard deviation σ minutes.

Given that 10% of the calls take longer than 22 minutes,

- (a) show that, to 3 significant figures, the value of σ is 4.68

(3)

- (b) Calculate the percentage of calls that take less than 13 minutes.

(1)

A supervisor in the call centre claims that the mean call time is less than 16 minutes. He collects data on his own call times.

- 20% of the supervisor's calls take more than 17 minutes to complete.
- 10% of the supervisor's calls take less than 8 minutes to complete.

Assuming that the time the supervisor takes to complete a call may be modelled by a normal distribution,

- (c) estimate the mean and the standard deviation of the time taken by the supervisor to complete a call.

(6)

- (d) State, giving a reason, whether or not the calculations in part (c) support the supervisor's claim.

(1)



Question 4 continued

Lined area for writing the answer to Question 4.

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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 11 marks)



5. A fast food company has a scratchcard competition. It has ordered scratchcards for the competition and requested that 45% of the scratchcards be winning scratchcards.

A random sample of 20 of the scratchcards is collected from each of 8 of the fast food company's stores.

- (a) Assuming that 45% of the scratchcards are winning scratchcards, calculate the probability that in at least 2 of the 8 stores, 12 or more of the scratchcards are winning scratchcards.

(5)

- (b) Write down 2 conditions under which the normal distribution may be used as an approximation to the binomial distribution.

(1)

A random sample of 300 of the scratchcards is taken. Assuming that 45% of all the scratchcards are winning scratchcards,

- (c) use a normal approximation to find the probability that at most 122 of these 300 scratchcards are winning scratchcards.

(4)

Given that 122 of the 300 scratchcards are winning scratchcards,

- (d) comment on whether or not there is evidence at the 5% significance level that the proportion of the company's scratchcards that are winning scratchcards is different from 45%

(1)

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Question 5 continued

Lined area for writing the answer to Question 5.

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Question 5 continued

Lined area for writing answers.

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(Total for Question 5 is 11 marks)

TOTAL FOR STATISTICS IS 50 MARKS



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Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Mock Paper – Set 2			
		Paper Reference 9MA0/31	
Mathematics			
Advanced			
Paper 31: Statistics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

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Advice

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Turn over ►

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1. The partially completed table below summarises the times taken by 120 job applicants to complete a task.

Time, t (minutes)	$5 < t \leq 7$	$7 < t \leq 10$	$10 < t \leq 14$	$14 < t \leq 18$	$18 < t \leq 30$
Frequency	10	23	51		

A histogram is drawn.

The bar representing the $5 < t \leq 7$ has a width of 1 cm and a height of 5 cm.

- (a) Given that the bar representing the group $14 < t \leq 18$ has a height of 4 cm, find the frequency of this group.

(2)

- (b) Showing your working, estimate the mean time taken by the 120 job applicants.

(3)

The lower quartile of the times is 9.6 minutes and the upper quartile of the times is 15.5 minutes.

For these data, an outlier is classified as any value greater than $Q_3 + 1.5 \times \text{IQR}$.

- (c) Showing your working, explain whether or not any of the times taken by these 120 job applicants might be classified as outliers.

(2)

Candidates with the fastest 5% of times for the task are given interviews.

- (d) Estimate the time taken by a job applicant, below which they might be given an interview.

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 9 marks)



2. A local sports centre has showers with two temperature settings, warm and hot.

On the warm setting, the water temperature may be modelled by a Normal distribution with mean 30 °C and standard deviation 2 °C

- (a) Using the model, find the probability that the next time the shower is used on the warm setting, the water temperature is

- (i) exactly 31°C
- (ii) more than 31°C

(2)

The sports centre manager thinks that a water temperature of more than 33 °C is too high for the warm setting.

She tests the water temperature on the warm setting on 5 randomly selected days.

Given that the probability of the water temperature being more than 33 °C is 0.0668

- (b) find the probability of the water temperature being more than 33°C

- (i) on only the first of these 5 days,

(2)

- (ii) on more than 1 of these 5 days.

(3)

On the hot setting, the water temperature may be modelled by a Normal distribution with standard deviation 1.5°C

The probability that the water temperature is more than 42°C is 0.0005

- (c) Find the mean water temperature on this setting, giving your answer to 1 decimal place.

(4)

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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 11 marks)



3. Sam is investigating the weather throughout the year in Hurn.
He uses the large data set to investigate the daily mean wind direction.

Sam believes that each cardinal wind direction is equally likely in Hurn.

(a) Assuming that Sam is correct,

- (i) state the probability that the cardinal wind direction in Hurn on a randomly selected day is NNE, (1)
- (ii) state the distribution that Sam should use to model the probability of each cardinal wind direction in Hurn on a randomly selected day. (1)

Sam decides to investigate the daily mean wind direction throughout the year.

(b) State a limitation of using the data for Hurn from the large data set as a sampling frame. (1)

(c) Explain how to use simple random sampling to select 36 days from a year. (2)

Sam defines the random variable X as the number of days out of 36 on which the daily mean wind direction in Hurn is between the bearings 135° and 225°

Sam collects data from 36 randomly selected days and finds that $x = 15$

Sam carries out a hypothesis test at the 10% level of significance.

- (d) Given that $H_0: p = 0.25$ and that the critical region is $\{X \leq 4 \cup X \geq 14\}$
- (i) state the alternative hypothesis, H_1
 - (ii) giving a reason for your answer, explain what Sam should conclude about the daily mean wind direction in Hurn. (3)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 8 marks)



4. A large number of cyclists take part in a cycling time trial. A random sample of these cyclists are selected and their times, in minutes, are summarised in the following statistics

$$\sum x = 1680 \quad \sum x^2 = 47654.4 \quad n = 60$$

- (a) Calculate, for this sample, the value of

- (i) the mean time, (1)

- (ii) the standard deviation of the times. (2)

Historically, the mean time for cyclists on this time trial has been 27 minutes and 30 seconds. Lucy is watching the time trial and believes that the mean time of cyclists in this time trial is greater than the mean time of cyclists in previous time trials.

The times of cyclists on this time trial are modelled by a Normal distribution with standard deviation 3 minutes.

- (b) Test, at the 5% level of significance, whether or not this sample provides evidence to support Lucy's belief. You should state your hypotheses and show your working clearly. (5)

Speedy Wheels cycling club entered its 5 fastest riders and 5 beginners to take part in the time trial.

The fastest 20% of the cyclists in the time trial are invited to compete in a race the following week.

- (c) (i) Explain, with specific reference to the parameter p , why the distribution $B(10, 0.2)$ might not be reasonable to model the number of these *Speedy Wheels* cycling club members who are invited to compete in the race. (2)
- (ii) Suggest how to improve the model for the number of these *Speedy Wheels* cycling club members invited to compete in the race. (1)



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Question 4 continued

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DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 11 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 11 marks)

TOTAL FOR STATISTICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Mock Paper Set 3			
		Paper Reference 9MA0/31	
Mathematics			
Advanced			
Paper 31: Statistics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Sam is playing a computer game.

When Sam earns a reward in the game, she randomly receives either a Silver reward or a Gold reward.

Each time that Sam earns a reward, the probability of receiving a Gold reward is 0.4

One day Sam plays the computer game and earns 11 rewards.

(a) Find the probability that she receives

(i) exactly 2 Gold rewards, (2)

(ii) at least 5 Gold rewards. (2)

In the next month Sam earns 300 rewards.

She decides to use a Normal distribution to estimate the probability that she will receive at least 135 Gold rewards.

(b) (i) Find the mean and variance of this Normal distribution. (2)

(ii) Estimate the probability that Sam will receive at least 135 Gold rewards. (2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)

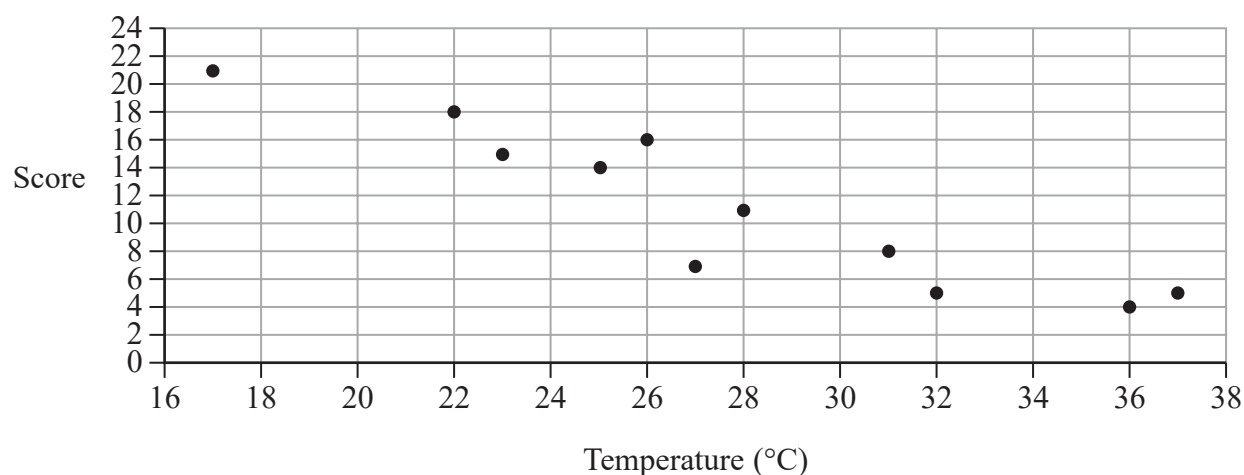


2. Xiang is investigating how room temperature affects a person's score in a task.

She gets Simon to complete the task 11 times at various controlled room temperatures, $x^{\circ}\text{C}$.

Xiang records the temperature, x , and Simon's score, y , where y is an integer.

The results are shown in the scatter diagram below.



- (a) Use the scatter diagram to find

- (i) the median score
- (ii) the range of the scores.

(2)

The temperature was increased each time Simon completed the task.

Xiang believes that as the room temperature increases, Simon's score will decrease.

Xiang calculates the product moment correlation coefficient from her data as -0.9286

- (b) Use this calculated value to carry out a suitable hypothesis test to investigate her belief at a 5% level of significance.
State clearly

- your hypotheses
- your critical value

(3)

Xiang is concerned that because Simon is repeating the same task his scores may improve.

- (c) Comment on how this concern may affect Xiang's conclusion to the test in part (b).

(1)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



3.

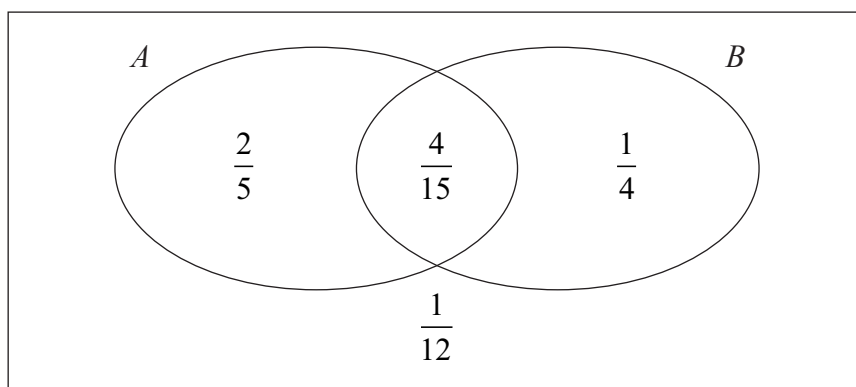


Figure 1

Figure 1 shows a Venn diagram with two events, A and B , and their associated probabilities.

- (a) Explain whether or not events A and B are independent.
Show your working clearly.

(3)

- (b) Find $P(B|A')$

(2)

- (c) Complete the tree diagram in Figure 2 by calculating the probabilities associated with each branch.

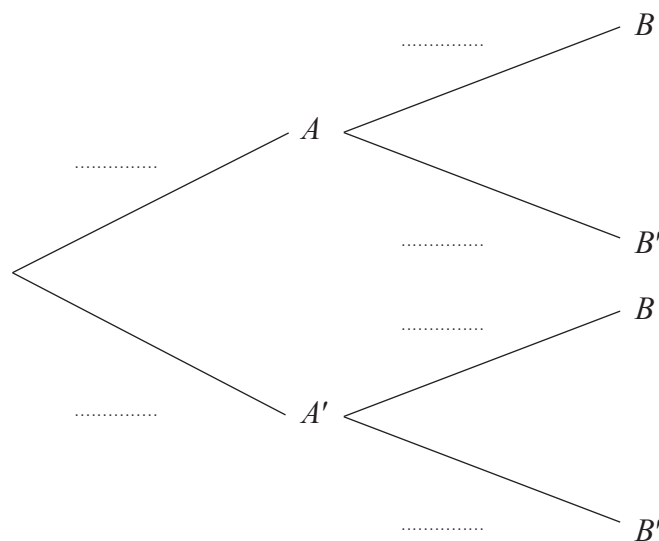


Figure 2

(4)

Turn over for a spare copy of Figure 2 if you need to redraw your tree diagram.



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

Only use this copy of Figure 2 if you need to redraw your tree diagram.

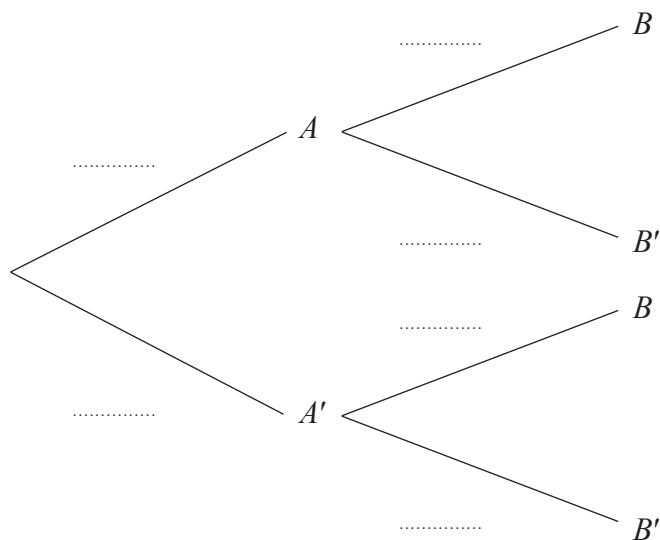


Figure 2

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 9 marks)



4. Kaiyang and Maggie are studying the pattern of rainfall where they live in Hurn.

Kaiyang decides to record whether or not it is raining at 6 a.m. for the next 10 days.

- (a) State the name of the sampling method Kaiyang is using. (1)

Kaiyang suggests that the binomial distribution with $n = 10$ would be a good model for the number of days in a 10-day period when it is raining at 6 a.m.

- (b) Explain why the binomial distribution might not be a good model in this situation. (1)

Maggie uses data from the large data set for Hurn in 2015.

She randomly selects 30 days from the large data set for Hurn in 2015 and for each day records the Daily Total Rainfall, x (mm)

Some of her 30 days have 'tr' recorded as the Daily Total Rainfall.

- (c) Using your knowledge of the large data set
- state what 'tr' means
 - suggest what Maggie could do with these 'tr' entries to clean her data set
 - explain what effect this action is likely to have on an estimate of the mean Daily Total Rainfall
- (3)

Maggie summarises her data in the grouped frequency table shown below.

Daily Total Rainfall, x (mm)	0	$0 < x \leq 1$	$1 < x \leq 3$	$3 < x \leq 7$	$7 < x \leq 15$
Frequency, f	7	10	2	7	4

- (d) Use this information to estimate the mean Daily Total Rainfall for these 30 days. (2)

- (e) Show that the estimated value of S_{xx} for the Daily Total Rainfall for these 30 days is 411.4 to 1 decimal place. (3)

Maggie defines an outlier as a value which is more than 3 standard deviations from the mean.

- (f) State, giving a reason, whether or not there could be any possible outliers in Maggie's data. (3)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 13 marks)



5. Kim and Tom are both learning to tune a violin.

Kim's teacher asks her to tune the *A-string* of her violin to the correct frequency in hertz (Hz).

When Kim tunes the *A-string*, its frequency may be modelled by a Normal distribution.

When Kim first starts learning, she tunes the *A-string* with a mean frequency of 443 Hz and a standard deviation of 6 Hz.

The correct frequency for the *A-string* is 440 Hz.

Find the probability that Kim tunes the *A-string*

- (a) lower than the correct frequency, (2)

- (b) more than 5 Hz away from the correct frequency. (2)

After practising for a month, Kim tunes the *A-string* with a standard deviation of 4.5 Hz.

She claims that the mean frequency when she tunes the *A-string* is now less than 443 Hz.

Kim's teacher asks Kim to tune the *A-string* 20 times and finds that the mean frequency is 442 Hz.

- (c) Test at the 5% level of significance whether or not there is evidence to support Kim's claim.

You should state your hypotheses and show your working clearly. (4)

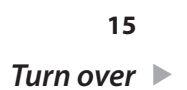
When Tom tunes the *A-string*, its frequency, T Hz, may be modelled by $T \sim N(\mu, \sigma^2)$

Given that $P(T < 438) = 0.2$ and that $P(T > 445) = 0.1$

- (d) find the value of μ and the value of σ , giving your answers, in Hz, to 1 decimal place. (6)



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Question 5 continued

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(Total for Question 5 is 14 marks)

TOTAL FOR STATISTICS IS 50 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Paper
reference

9MA0/31

Mathematics

Advanced

PAPER 31: Statistics

Mock Set 4

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 50. There are 6 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

1. Jen is using the large data set to investigate the relationship between Daily Maximum Relative Humidity and Daily Mean Visibility.

Using the data for Camborne, Jen takes as her sample the first 8 days of October 2015

Jen's data, without units, is given in the table below.

Daily Maximum Relative Humidity	93	92	97	98	100	99	91	96
Daily Mean Visibility	1100	1000	700	700	300	700	2300	1600

The Daily Mean Visibility for 1st October was recorded as 1100

- (a) Give an interpretation of this value, including any units.

(2)

The product moment correlation coefficient for Jen's data, to 3 decimal places, is in this list

0.316 -0.256 -0.484 -0.757 -1.035

One of these values **cannot** be a product moment correlation coefficient.

- (b) Explain which value.

(1)

- (c) Use your calculator to identify from the list the correct product moment correlation coefficient for Jen's data.

(1)

Jen believes that the Daily Maximum Relative Humidity each day can be used to predict the Daily Mean Visibility.

- (d) Carry out a suitable test, at the 5% significance level, to show why the data may support Jen's belief.

State clearly

- your hypotheses
- your critical value

(3)

- (e) Give **two** reasons why Jen's belief may still not be correct.

(2)



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Question 1 continued.

Lined area for writing the answer to Question 1.



Question 1 continued.

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Question 1 continued.

Lined area for writing answers.

(Total for Question 1 is 9 marks)



2. When a piece of buttered toast is dropped, the probability that it lands butter side up is p

Two identical pieces of buttered toast are dropped independently in the same way.

The random variable T represents the number of pieces that land butter side up, so that T can take the value 0, 1 or 2

- (a) Find $P(T = 0)$, giving your answer in terms of p (1)

- (b) Show that the distribution of T is not discrete uniform. (3)



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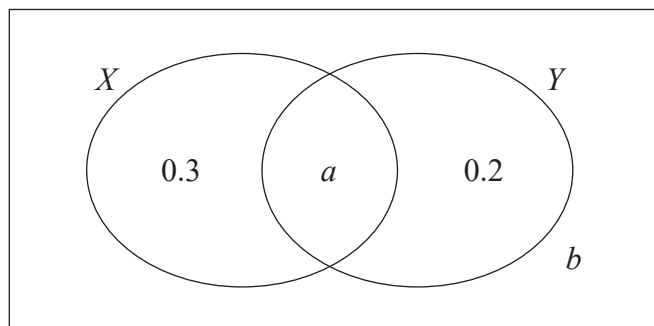
Question 2 continued.

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(Total for Question 2 is 4 marks)



3. The Venn diagram shows probabilities related to two events, X and Y



- (a) (i) State the value of $P(Y \cap X')$
 (ii) Find $P(Y|X')$ giving your answer in terms of b (3)

Given that $P(X|Y) = P(Y|X')$

- (b) show that $ab = 0.04$ (3)

Given also that $a > b$

- (c) find the value of a and the value of b
 Show your working clearly. (5)

- (d) Hence show that X and Y are not independent. (2)



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Question 3 continued.

Lined area for writing the answer to Question 3.



Question 3 continued.

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Question 3 continued.

Handwriting practice area with horizontal lines.

(Total for Question 3 is 13 marks)



4. In a game of *Sixes*, each player's turn involves rolling 6 identical dice.
If the player gets a six on fewer than 3 dice, the player does not score points.

Assuming that the dice are fair, find the probability that in one turn a player

- (a) (i) gets a six on exactly 3 dice
(ii) gets a six on at least 3 dice

(3)

Ali and four of his friends play *Sixes* together and have one turn each.
Two of the 5 players score points.

Ali claims that this suggests the dice are biased towards rolling a six.

- (b) Carry out a suitable test to investigate Ali's claim.

You should

- state your hypotheses clearly
- use a 5% level of significance
- state the p -value for the test

(4)



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Question 4 continued.

Lined area for writing the answer to Question 4.

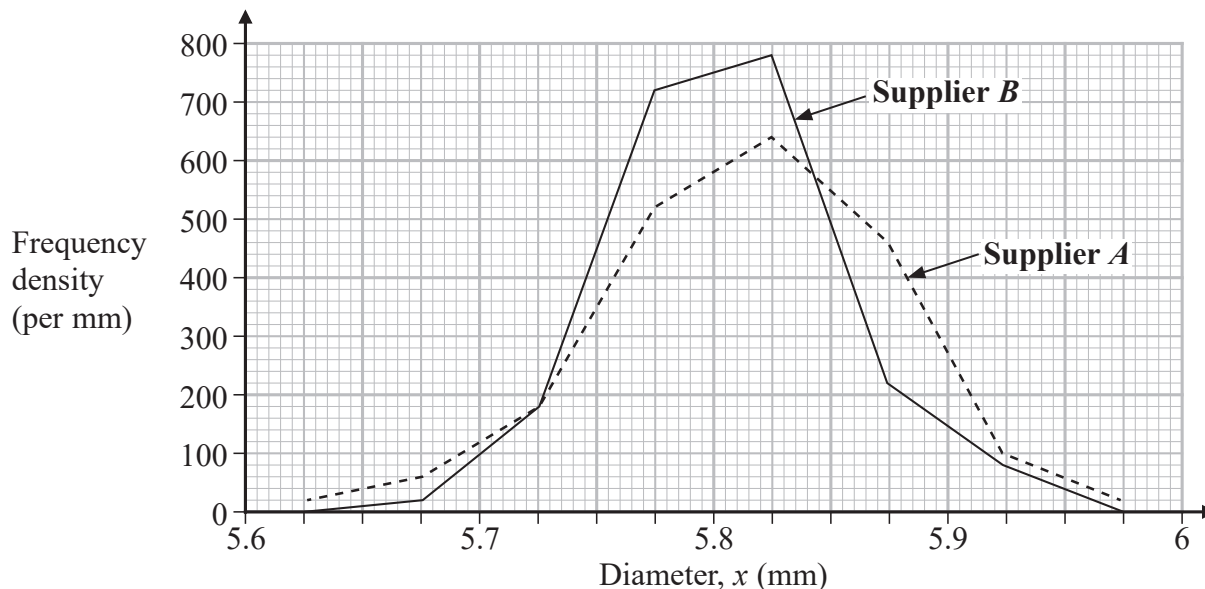
(Total for Question 4 is 7 marks)



5. An engineering company buys components with a specified diameter of 5.8 mm. Only components with a diameter between 5.75 mm and 5.85 mm are suitable for use.

A random sample of 100 of these components is taken from each of supplier *A* and supplier *B*.

The diameters are grouped into 0.05 mm intervals and the results are shown in the frequency polygons below.



Sam believes that supplier *B* will supply components with a more consistent diameter.

- (a) Explain how the frequency polygons support Sam's belief. (1)

The modal class for each supplier is $5.80 \leq x < 5.85$

- (b) Determine how many more components are in this class for supplier *B* than for supplier *A* in these samples. (2)

Sam also finds these summary statistics for the sample from supplier *B*

$$\sum x^2 = 3371.1975 \quad \sigma = 0.0483632$$

- (c) Determine the mean for this sample. (2)

The diameter of components, D mm, from supplier *B* may be modelled by

$$D \sim N(5.81, 0.048^2)$$

Any component with a diameter less than 5.75 mm is not used.

- (d) Using the model, find the probability that a component randomly selected from supplier *B* is not used. (1)

Any component with a diameter more than 5.85 mm is machined down by the company and is then suitable for use.

- (e) In a random sample of 40 components from supplier *B* that are suitable for use, estimate how many had been machined down by the company. (4)



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Question 5 continued.

Lined area for writing the answer to Question 5.



Question 5 continued.

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Question 5 continued.

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(Total for Question 5 is 10 marks)



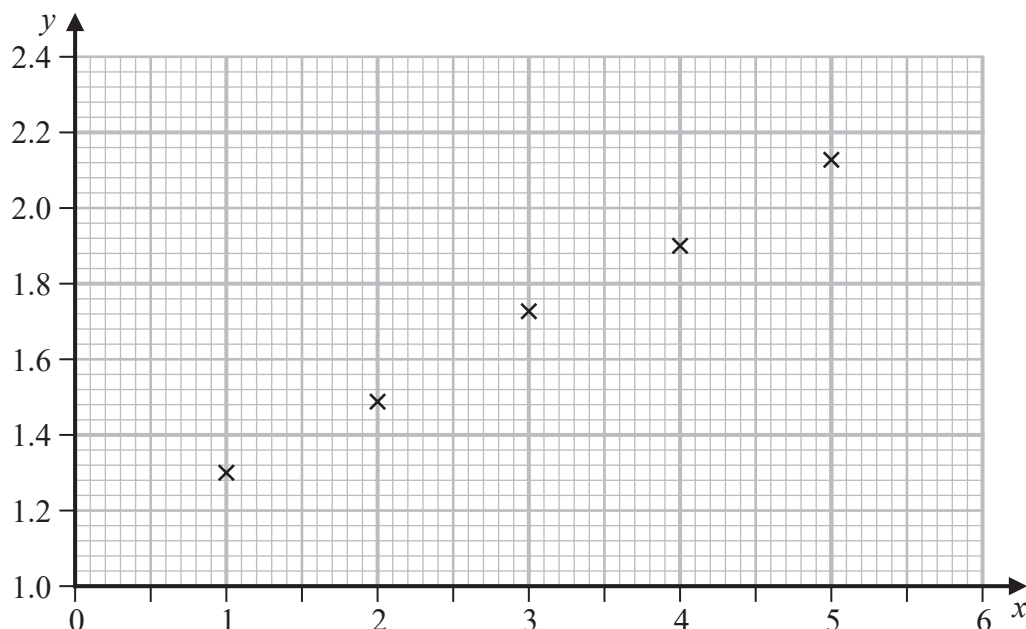
6. Roberta started a business five years ago.

She believes that the number of customers, c , is growing exponentially.

She produced the graph below by coding her data such that

x = number of years since the business started

$y = \log_{10} c$



Roberta found the regression line for this graph to be $y = 1.10 + 0.204x$

- (a) (i) Explain how the graph supports Roberta's belief of exponential growth.
- (ii) Find the relationship between the number of customers and number of years since the business started, in the form $c = ab^x$

(5)

Roberta claims that after 6 years she will have more than 200 customers.

- (b) Show that Roberta's model supports this claim.

(1)

- (c) Comment on the reliability of using Roberta's model in your answer to part (b). You must give a reason for your answer.

(1)



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Question 6 continued.

Lined area for writing the answer to Question 6.



Question 6 continued.

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(Total for Question 6 is 7 marks)

TOTAL FOR STATISTICS IS 50 MARKS



Write your name here

Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced

Paper 3: Statistics and Mechanics

Sample Assessment Material for first teaching September 2017

Time: 2 hours

Paper Reference

9MA0/03

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer cross it out and put your new answer and any working out underneath.

Turn over ►

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SECTION A: STATISTICS

Answer ALL questions. Write your answers in the spaces provided.

1. The number of hours of sunshine each day, y , for the month of July at Heathrow are summarised in the table below.

Hours	$0 \leq y < 5$	$5 \leq y < 8$	$8 \leq y < 11$	$11 \leq y < 12$	$12 \leq y < 14$
Frequency	12	6	8	3	2

A histogram was drawn to represent these data. The $8 \leq y < 11$ group was represented by a bar of width 1.5 cm and height 8 cm.

- (a) Find the width and the height of the $0 \leq y < 5$ group. (3)

- (b) Use your calculator to estimate the mean and the standard deviation of the number of hours of sunshine each day, for the month of July at Heathrow.
Give your answers to 3 significant figures. (3)

The mean and standard deviation for the number of hours of daily sunshine for the same month in Hurn are 5.98 hours and 4.12 hours respectively.

Thomas believes that the further south you are the more consistent should be the number of hours of daily sunshine.

- (c) State, giving a reason, whether or not the calculations in part (b) support Thomas' belief. (2)

- (d) Estimate the number of days in July at Heathrow where the number of hours of sunshine is more than 1 standard deviation above the mean. (2)

Helen models the number of hours of sunshine each day, for the month of July at Heathrow by $N(6.6, 3.7^2)$.

- (e) Use Helen's model to predict the number of days in July at Heathrow when the number of hours of sunshine is more than 1 standard deviation above the mean. (2)

- (f) Use your answers to part (d) and part (e) to comment on the suitability of Helen's model. (1)

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Question 1 continued

(Total for Question 1 is 13 marks)

2. A meteorologist believes that there is a relationship between the daily mean windspeed, w kn, and the daily mean temperature, t °C. A random sample of 9 consecutive days is taken from past records from a town in the UK in July and the relevant data is given in the table below.

t	13.3	16.2	15.7	16.6	16.3	16.4	19.3	17.1	13.2
w	7	11	8	11	13	8	15	10	11

The meteorologist calculated the product moment correlation coefficient for the 9 days and obtained $r = 0.609$

- (a) Explain why a linear regression model based on these data is unreliable on a day when the mean temperature is 24 °C (1)
- (b) State what is measured by the product moment correlation coefficient. (1)
- (c) Stating your hypotheses clearly test, at the 5% significance level, whether or not the product moment correlation coefficient for the population is greater than zero. (3)

Using the same 9 days a location from the large data set gave $\bar{t} = 27.2$ and $\bar{w} = 3.5$

- (d) Using your knowledge of the large data set, suggest, giving your reason, the location that gave rise to these statistics. (1)

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Question 2 continued

(Total for Question 2 is 6 marks)

3. A machine cuts strips of metal to length L cm, where L is normally distributed with standard deviation 0.5 cm.

Strips with length either less than 49 cm or greater than 50.75 cm **cannot** be used.

Given that 2.5% of the cut lengths exceed 50.98 cm,

- (a) find the probability that a randomly chosen strip of metal **can** be used.

(5)

Ten strips of metal are selected at random.

- (b) Find the probability fewer than 4 of these strips **cannot** be used.

(2)

A second machine cuts strips of metal of length X cm, where X is normally distributed with standard deviation 0.6 cm

A random sample of 15 strips cut by this second machine was found to have a mean length of 50.4 cm

- (c) Stating your hypotheses clearly and using a 1% level of significance, test whether or not the mean length of all the strips, cut by the second machine, is greater than 50.1 cm

(5)

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Question 3 continued

(Total for Question 3 is 12 marks)

4. Given that

$$P(A) = 0.35 \quad P(B) = 0.45 \quad \text{and} \quad P(A \cap B) = 0.13$$

find

(a) $P(A' \mid B')$ (2)

(b) Explain why the events A and B are not independent. (1)

The event C has $P(C) = 0.20$

The events A and C are mutually exclusive and the events B and C are statistically independent.

(c) Draw a Venn diagram to illustrate the events A , B and C , giving the probabilities for each region. (5)

(d) Find $P([B \cup C]')$ (2)

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Question 4 continued

(Total for Question 4 is 10 marks)

5. A company sells seeds and claims that 55% of its pea seeds germinate.

- (a) Write down a reason why the company should not justify their claim by testing all the pea seeds they produce.

(1)

A random selection of the pea seeds is planted in 10 trays with 24 seeds in each tray.

- (b) Assuming that the company's claim is correct, calculate the probability that in at least half of the trays 15 or more of the seeds germinate.

(3)

- (c) Write down two conditions under which the normal distribution may be used as an approximation to the binomial distribution.

(1)

A random sample of 240 pea seeds was planted and 150 of these seeds germinated.

- (d) Assuming that the company's claim is correct, use a normal approximation to find the probability that at least 150 pea seeds germinate.

(3)

- (e) Using your answer to part (d), comment on whether or not the proportion of the company's pea seeds that germinate is different from the company's claim of 55%

(1)

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TOTAL FOR SECTION A IS 50 MARKS

Write your name here

Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced Subsidiary

Paper 3: Statistics and Mechanics

Sample Assessment Material for first teaching September 2017

Time: 2 hours

Paper Reference

9MA0/03

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Turn over ►

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SECTION A: STATISTICS

Answer ALL questions. Write your answers in the spaces provided.

1. *Kaff coffee* is sold in packets. A seller measures the masses of the contents of a random sample of 90 packets of *Kaff coffee* from her stock. The results are shown in the table below.

Mass w (g)	Midpoint y (g)	Frequency (f)
$240 \leq w < 245$	242.5	8
$245 \leq w < 248$	246.5	15
$248 \leq w < 252$	250	35
$252 \leq w < 255$	253.5	23
$255 \leq w < 260$	257.5	9

(You may use $\sum fy^2 = 5\,644\,171.75$)

A histogram is drawn and the class $245 \leq w < 248$ is represented by a rectangle of width 1.2 cm and height 10 cm.

- (a) Calculate the width and the height of the rectangle representing the class $255 \leq w < 260$ (3)
- (b) Use linear interpolation to estimate the median mass of the contents of a packet of *Kaff coffee* to 1 decimal place. (2)
- (c) Estimate the mean and the standard deviation of the mass of the contents of a packet of *Kaff coffee* to 1 decimal place. (3)

The seller claims that the mean mass of the contents of the packets is more than the stated mass.

Given that the stated mass of the contents of a packet of *Kaff coffee* is 250 g and the actual standard deviation of the contents of a packet of *Kaff coffee* is 4 g,

- (d) test, using a 5% level of significance, whether or not the seller's claim is justified. State your hypotheses clearly. (You may assume that the mass of the contents of a packet is normally distributed.) (5)
- (e) Using your answers to parts (b) and (c), comment on the assumption that the mass of the contents of a packet is normally distributed. (1)



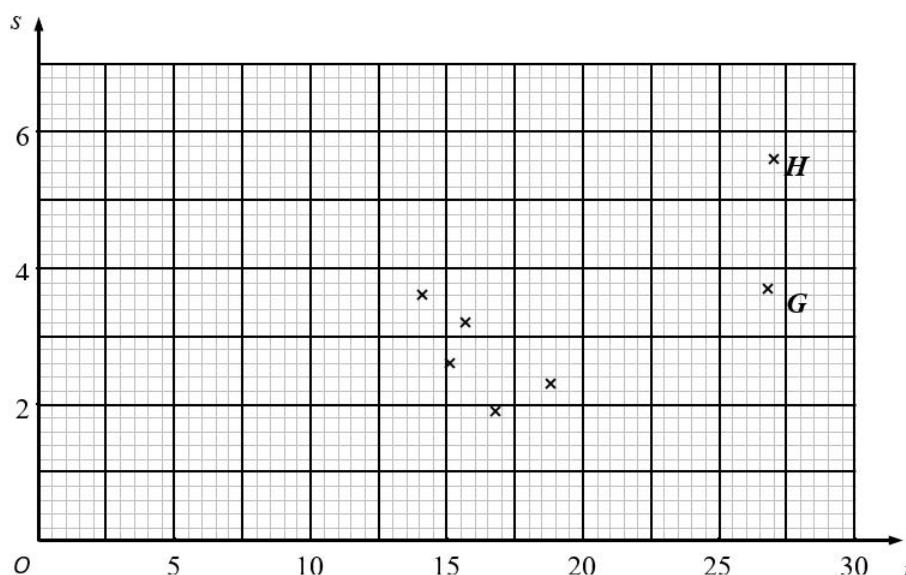
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Question 1 continued

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(Total for Question 1 is 14 marks)

2. A researcher believes that there is a linear relationship between daily mean temperature and daily total rainfall. The 7 places in the northern hemisphere from the large data set are used. The mean of the daily mean temperatures, t °C, and the mean of the daily total rainfall, s mm, for the month of July in 2015 are shown on the scatter diagram below.



- (a) With reference to the scatter diagram, explain why a linear regression model may not be suitable for the relationship between t and s .

(1)

The researcher calculated the product moment correlation coefficient for the 7 places and obtained $r = 0.658$

- (b) Stating your hypotheses clearly, test at the 10% level of significance, whether or not the product moment correlation coefficient for the population is greater than zero.

(3)

- (c) Using your knowledge of the large data set, suggest the names of the 2 places labelled G and H .

(1)

- (d) Using your knowledge from the large data set, and with reference to the locations of the 2 places labelled G and H , give a reason why these places have the highest temperatures in July.

(1)

- (e) Suggest how you could make better use of the large data set to investigate the relationship between daily mean temperature and daily total rainfall.

(1)

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3. For a particular type of bulb, 36% grow into plants with blue flowers and the remainder grow into plants with white flowers. Bulbs are sold in mixed bags of 40.

Russell selects a random sample of 5 bags of bulbs.

- (a) Find the probability that fewer than 2 of these bags will contain more bulbs that grow into plants with blue flowers than grow into plants with white flowers

Maggie takes a random sample of n bulbs.

Using a normal approximation, the probability that more than 244 of these n bulbs will grow into blue flowers is 0.0521 to 4 decimal places.

- (b) Find the value of n . (6)

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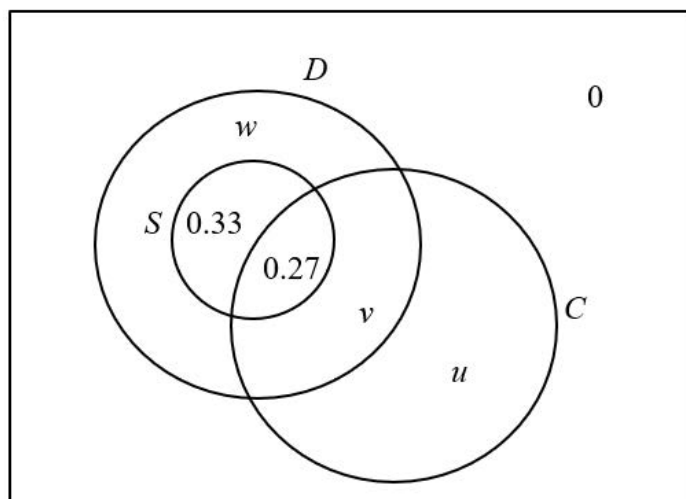
4. The Venn diagram shows the probabilities of students' lunch boxes containing a drink, sandwiches and a chocolate bar.

D is the event that a lunch box contains a drink,

S is the event that a lunch box contains sandwiches,

C is the event that a lunch box contains a chocolate bar,

u , v and w are probabilities.



- (a) Write down $P(S \cap D')$ (1)

One day, 80 students each bring in a lunch box.

Given that all 80 lunch boxes contain sandwiches and a drink,

- (b) estimate how many of these 80 lunch boxes will contain a chocolate bar. (3)

Given that the events S and C are independent and that $P(D|C) = \frac{14}{15}$

- (c) calculate the value of u , the value of v and the value of w . (7)

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5. The lifetimes of batteries sold by company X are normally distributed, with mean 150 hours and standard deviation 25 hours.

A box contains 12 batteries from company X .

- (a) Find the expected number of these batteries that have a lifetime of more than 160 hours. (3)

The lifetimes of batteries sold by company Y are normally distributed, with mean 160 hours and 80% of these batteries have a lifetime of less than 180 hours.

- (b) Find the standard deviation of the lifetimes of batteries from company Y . (3)

Both companies sell their batteries for the same price.

- (c) State which company you would recommend. Give reasons for your answer. (2)

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Question 5 continued

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(Total for Question 5 is 8 marks)

TOTAL FOR SECTION A IS 50 MARKS